CS398: CODING WITH AI

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Goal of this Class

Understand how to work with and get the best out of AI-based developer tools to develop Software that you can ship in an Enterprise.

Setting Expectations: What this class is not about?

- We are not going to see how to build or improve the AI models.
- We are not going to learn how to build software applications that use AI models – multi agent systems.
- We are not going to be vibe coding.
- We are not going to chase after the fad of the month.

Setting Expectations: What this class is about?

- Learning how to get the best out of AI-based dev tools to build enterprise grade software from research and other practitioner experiences.
- Learn practices that will stand the test of time and will apply no matter which tool or model you use.
- Lots of in-class programming, unlike many other undergrad courses: We will learn best practices and then immediately apply them.
- Attendance Compulsory
- No Homework.

Course Structure (Weeks 3-12)

- Each week in on one topic in the Software Engineering life cycle.
- Each topic is covered in 2 classes.
- Each class you will learn about 3 Guidelines/Best practices.
- Each guideline will be followed up with an exercise to practice implementing the guideline.
- We will share the solution and system/user prompts.
  - Manually compare against your submission.
  - Use the system/user prompt to get qualitative feedback from a foundation model.

Get Access to one of the following tools.

OpenAI Codex / Claude Code / Cursor / GitHub Copilot / Amazon Kiro

Cost: 25-30 CAD/Month
Amazon Kiro is free for UWaterloo students for a year.

Laptop is required for this class. If you do not have one, email me and we can set you up with a loaner.

Let's walk through the Syllabus and Schedule

https://cs.uwaterloo.ca/~m2nagapp/courses/CS398/1265/index.html

Do I really need to do this course?

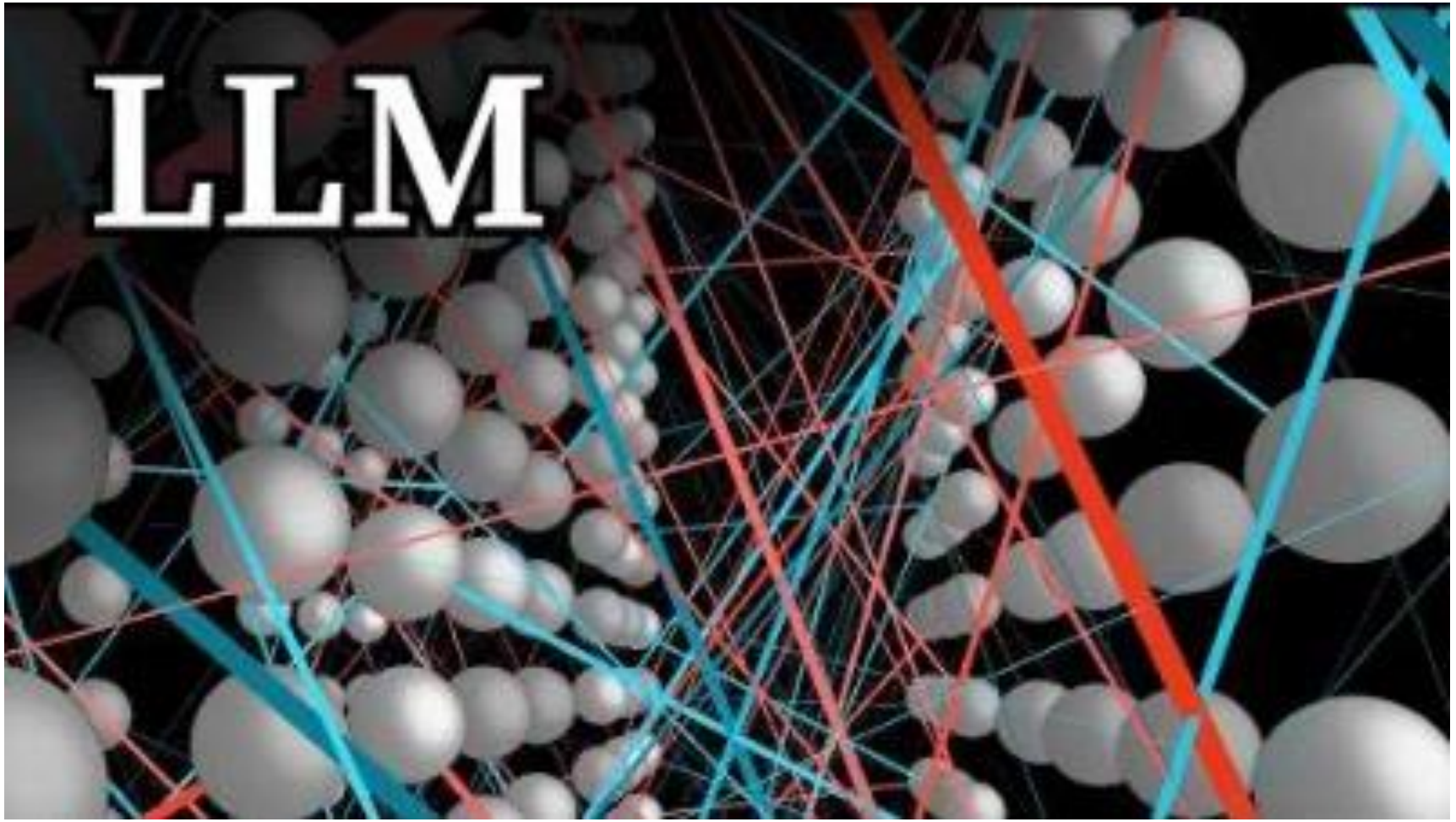
Dario Amodei - CEO of Anthropic (March 2025): “I think we’ll be there in three to six months—where AI is writing 90 percent of the code. And then in twelve months, we may be in a world where AI is writing essentially all of the code”.

All Generative AI Company CEOs have said something to this effect.

Do I really need to do this course? While AI might write the code...

- Coding was never the bottleneck in building software.
- Users still want high quality, reliable, secure and performant software.
- You are still responsible for the code.

Large Language Models explained briefly



Large Language Models and SE

- Software is still built for humans and humans do not know what they exactly want.
- Even if they know what they want, they cannot precisely express it in natural language.
- Given that, they are not going to replace you anytime soon. And a course like this is to help you use this powerful tool to your advantage.

Key Takeaway: Do not let AI make a choice for you.

- There is no one way to optimize on a choice.
- AI does not know which choice is important for you at this time.
- Use AI to enumerate the choices and their pros/cons. But make and own the decision.
- If you do not know enough to make a decision, then take the time to learn about it and then make the decision.

CS398: First offering of the course

- While I am familiar with the material, the delivery as a course is new.
- Your feedback throughout the course is very helpful!
  - On the class structure
  - Any mistakes in the material
  - How helpful the material is in your personal/professional work.

How to contact us?

- Email me or the TAs with your questions/for office hours.
- Include CS398 in the subject line for a reply.
- We will use Learn for all the material and submissions.
- Any changes will be posted here at the top of the page: https://cs.uwaterloo.ca/~m2nagapp/courses/CS398/1265/index.html